Ex. NO 7 (CY)

Date :- 21.10.25

Implementation of Subnetting in CISCO PACKET TRACER simulator

a) Design Multiple Subnets with a suitable Number of Host

Aim:

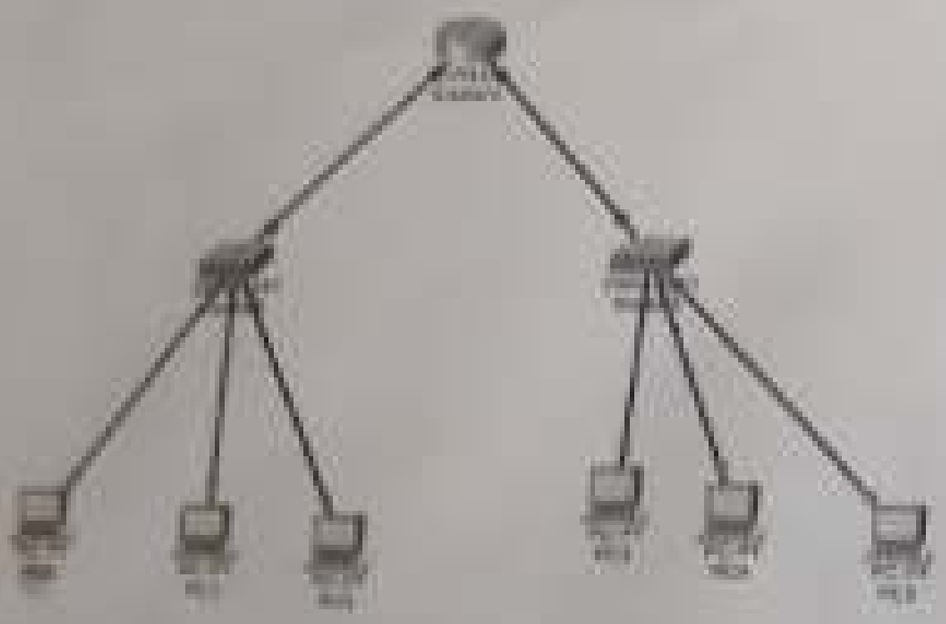
To Design Multiple Subnets with a suitable number of hosts using Subnetting techniques in CISCO Packet Tracer.

Steps :-

1. Open Cisco Packet Tracer and create a new project.
2. Design the network topology by placing the required network devices
3. Select a suitable network address for the experiment.
4. Calculate the number of Subnets required.
5. Determine the no. of host needed in each subnet.
6. Calculate the subnet mask using subnetting techniques
7. Divide the network into multiple Subnets
8. Assign a separate network address to each subnet.
9. Verify the subnet calculations and host ranges
10. Save the subnet topology.

Result :-

Thus, multiple Subnet were successfully designed according to the required no. of hosts.



PC0

Physical Config Desktop Custom Interface

IP Configuration

IP Configuration

DHCP Static

IP Address 192.168.1.11

Subnet Mask 255.255.255.0

Default Gateway 192.168.1.1

DNS Server

IPv6 Configuration

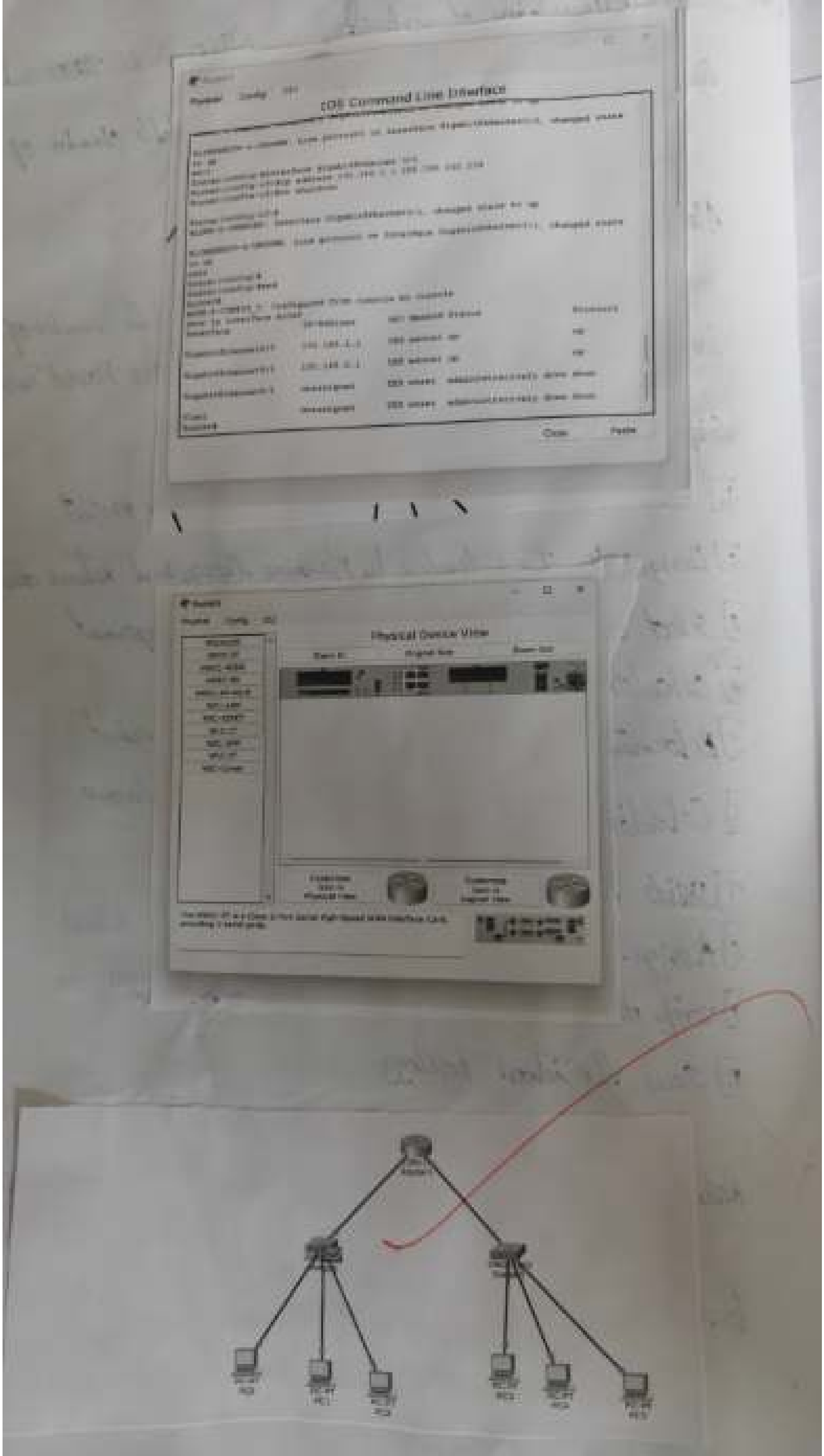
DHCP Auto Config Static

IPv6 Address /

Link Local Address FE80::2D0:BAFF:FE9C:D804

IPv6 Gateway

IPv6 DNS Server


CDS Command Line Interface
Physical Device View

b] Assign Static IP Addresses Across All Subnets and connect the Subnets Through a Router.

Aim:

To assign static IP address to all devices in different Subnet and establish communication between the subnet through a router.

Steps:

1] Connect a router in the switches using appropriate cables

2] Configure the router interfaces.

3] Assign an IP address and Subnet mask to each other router

4] Configure the switches

5] Assign static IP address to all computer.

6] Configure the Subnet mask for each device

7] Assign the appropriate default gateway to each computer

8] Verify the IP configuration of all devices.

9] Save the router configuration.

10] Verify the all devices are properly connected.

Result:

Thus, static IP addresses were assigned successfully and communication between the subnets was established through the router.

Command Prompt

Copy Paste

c] Simulate Packet Transmission Across the subnets and observe the Result.

d] When Subnets Are Connected through a Router

Aim:

To Analyze Packet Transmission between different subnet when a router is used for communication

Theory

1] Open Simulation Mode in Cisco Packet Tracer

2] Select a source device from one Subnet.

3] Select a destination device from another subnet.

4] Examine a packet using the Simple PDU tool

5] Observe the movement of the packet through the network

6] Analyze how the router receives the packet.

7] Observe the forwarding process performed by the router.

8] Verify the packet information between different subnet.

9] Use the ping command to test Connectivity.

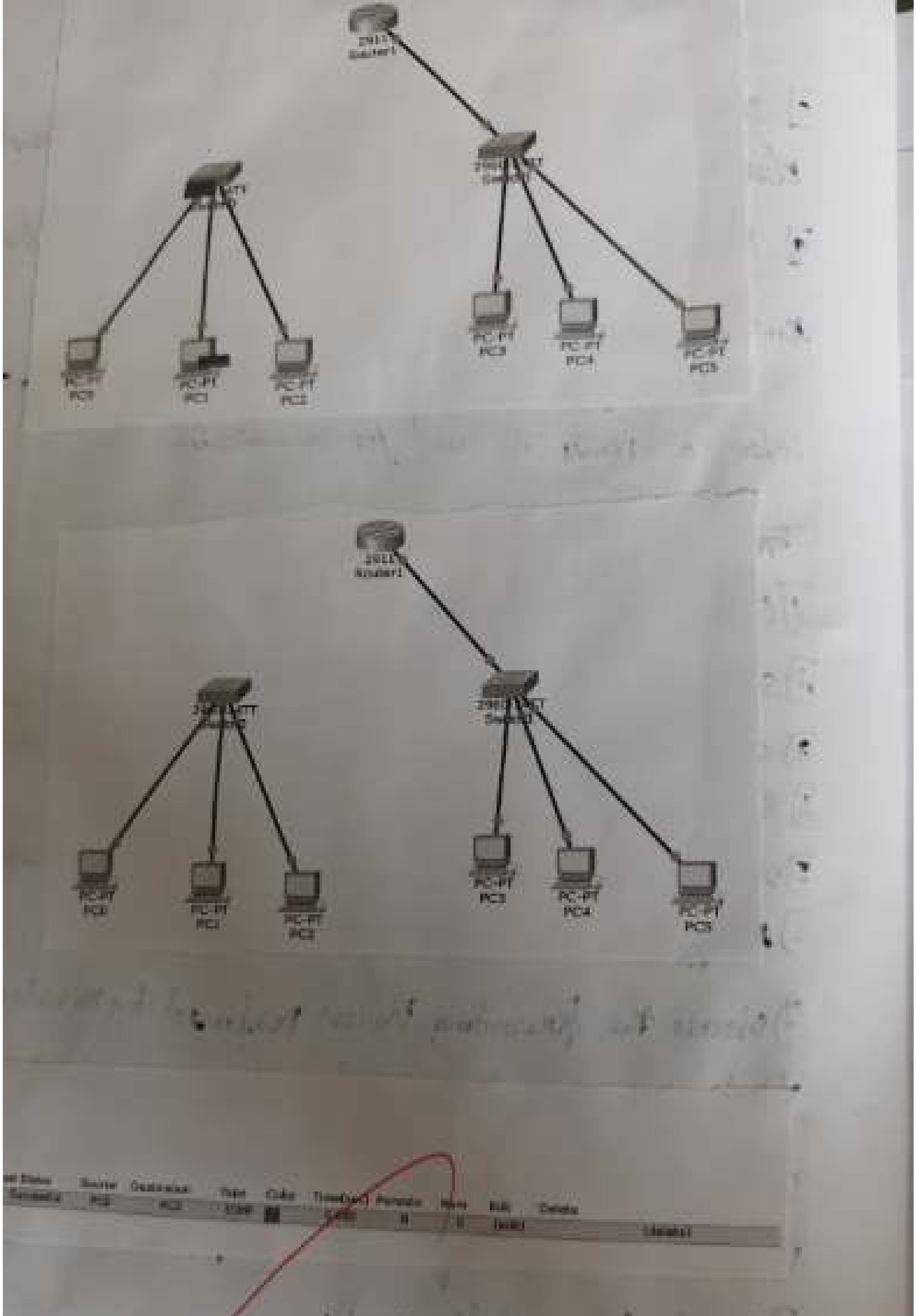
10] Observe the successful delivery of packets.

11] Record the simulation results.

12] Analyze the routing process.

Result:

Thus, Packets were transmitted successfully between different subnets through the router.


PC-PT
PC0
PC-PT
PC1
PC-PT
PC2
PC-PT
PC3
PC-PT
PC4
PC-PT
PC5

5] When Subnets are Not Connected Through a Router.

Aim:

To Analyse Packet transmission between different subnet when a router is not used for communication.

Steps:

1) Remove the router from the network topology
2) Open simulation mode.
3) Select device belonging to different subnets
4) Transmit packets using the simple PDU tool
5) Observe the packet transmission
6) Perform a ping test between devices located in different subnets.
7) Analyze the packet flows
8) Observe the simulator results
9) Identify the point at which packet transmission fails.
10) Compare the result with the network that uses a router
11) Record the observation
12) Analyze the implemented of a router in inter network communication.

Result:

Thus, Communication between different subnets failed because a router was not available to forward the packet